A Hypothesis on Consciousness as a Primary System of Brain Activity

Abstract

Consciousness is commonly discussed as a phenomenon associated with brain activity, but its fundamental nature remains unresolved. This paper proposes a hypothesis in which consciousness is not merely a by-product of neural activity but may represent a primary system that observes experience and potentially influences or activates the biological brain.

The hypothesis proposes that the brain continuously generates and processes information in the form of thoughts, sensations, memories, decisions, and other mental states. Some information is acted upon, some is stored in memory, and some is discarded. Consciousness, in this model, is distinguished from these individual contents. It is proposed as the system through which experience is observed.

A further hypothesis is that consciousness may provide an activating or organizing influence on brain activity, analogous conceptually to a power source or primary control system. This does not mean that consciousness has been demonstrated to be a physical form of energy. Rather, it proposes that if consciousness can be shown to produce measurable physical effects on the brain that cannot be explained by known neural mechanisms alone, consciousness may need to be considered a fundamental component of brain function.

This paper also examines sleep, memory, language, and subjective experience as possible observations relevant to the hypothesis. Particular attention is given to the distinction between consciousness itself and the information that appears within consciousness.

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1. Introduction

The human brain is one of the most complex physical systems known. It receives sensory information, processes signals, stores memories, generates thoughts, controls movement, and regulates numerous biological functions.

However, one question remains difficult:

Why does physical brain activity produce subjective experience?

A person does not merely process information. A person experiences information. There is an apparent internal perspective from which sensations, thoughts, memories, and surroundings are experienced.

This paper develops a personal theoretical hypothesis:

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Abstract and central proposal

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Claim: Consciousness may be an observing system distinct from individual thoughts, sensations and memories.

Experiment A — Separate awareness from reportable content. In a supervised study with consenting adult volunteers, present simple visual or auditory stimuli and collect both a direct awareness report and a separate description of what was experienced. Compare trials with similar sensory input but different reported awareness. Record response accuracy, confidence and timing. A result showing that awareness reports and specific content reports can vary independently would support the usefulness of distinguishing them conceptually; it would not establish a separate entity or system. A failure to obtain reliable distinctions would weaken this operational distinction.

Claim: Consciousness may also influence or activate brain activity.

Experiment B — Test a preregistered neural prediction. Before collecting data, specify a measurable brain signal and a quantitative prediction that differs between (1) a conventional neural-processing model and (2) the proposed additional-influence model. In an approved adult study, record EEG alongside stimulus timing, eye movements, behaviour and awareness reports. Test the models on held-out data. A reproducible effect beyond the conventional model would justify further investigation, not prove that consciousness caused it; hidden neural variables, measurement error and model misspecification remain alternatives. If the additional model makes no distinct prediction, the claim is not yet experimentally testable.

Claim: Thoughts and mental contents may be generated, selected, stored or discarded by brain processes.

Experiment C — Track information outcomes. Use a computer-based memory task with neutral images or symbols. Measure which items guide a response, which are recalled later and which are not accessible at test. Compare outcomes across repeated trials. This can test the information-processing description, but it cannot show that consciousness is a separate system.

Ethics and safety: Human studies require informed consent, privacy protection, qualified adult supervision and review by an appropriate research ethics committee. These proposals are for adult volunteers in approved settings. Do not use sleep deprivation, drugs, anaesthesia, brain stimulation or invasive procedures as do-it-yourself experiments.

> **Consciousness may be an observing and potentially activating system that interacts with the biological brain rather than being identical to the brain itself.**

The purpose of this paper is not to claim that this hypothesis has been proven. Its purpose is to formulate the idea precisely, identify its predictions, examine possible contradictions, and propose ways in which it could potentially be investigated scientifically.

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2. Definition of Consciousness

For this hypothesis, consciousness is defined as:

> **The capacity for subjective experience and observation of internal or external information.**

This definition intentionally separates consciousness from the particular information being experienced.

For example, a thought is not necessarily consciousness itself. A memory is not necessarily consciousness itself. A word is not consciousness itself.

Instead, these may be considered **contents of conscious experience**.

This leads to an important distinction:

Consciousness = the capacity or system associated with experiencing

Mental contents = information experienced by that system

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3. Proposed Properties of Consciousness

The following properties form the basis of the proposed model.

3.1 Consciousness observes

Conscious experience appears to contain an observing perspective. Visual information, sounds, sensations, thoughts, and memories can become objects of experience.

The hypothesis therefore treats consciousness as an **observer of mental and sensory information**.

This does not necessarily imply that consciousness is a separate physical observer located somewhere inside the brain. That remains an open question.

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Definition of consciousness; observation; distinction between awareness and mental contents

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Claim: Consciousness is the capacity for subjective experience of internal or external information.

Experiment D — Operationalise reports of experience. Ask adult participants to complete brief perception trials and provide immediate awareness ratings, confidence ratings and content descriptions. Use objective discrimination accuracy as a separate measure. Examine whether awareness ratings, discrimination and content descriptions converge or diverge. The experiment tests measurable indicators associated with experience; self-report is imperfect and is not a direct meter of consciousness.

Claim: A thought, memory or word is not identical to consciousness itself.

Experiment E — Compare awareness with content. Present the same type of stimulus under conditions that vary task demands, then collect (a) whether anything was experienced, (b) what was experienced and (c) whether it can be recalled later. If these measures dissociate, it supports treating them as different variables. It does not establish that they are separate substances or independent systems.

Claim: Consciousness observes sensory and mental information.

Experiment F — Compare external and internally generated content. In a safe, quiet computer task, compare awareness reports for a visible image, a remembered image and a spontaneous thought. Ask participants to classify the source after each trial. Analyse accuracy and confidence. This investigates how people report awareness of different content sources; it cannot locate an internal observer.

Ethics and safety: Human studies require informed consent, privacy protection, qualified adult supervision and review by an appropriate research ethics committee. These proposals are for adult volunteers in approved settings. Do not use sleep deprivation, drugs, anaesthesia, brain stimulation or invasive procedures as do-it-yourself experiments.

3.2 Consciousness does not inherently require language

Human beings frequently use language to describe thoughts, but conscious experience does not necessarily require language.

For example, a person can experience:

- a visual scene,
- a sound,
- a physical sensation,
- recognition of an object,
- an emotional state,

without necessarily converting the experience into spoken or written words.

Therefore, language may be considered a **tool used by the brain for representing and communicating information**, rather than a fundamental requirement for consciousness.

3.3 Consciousness may exist with minimal mental content

The hypothesis also considers whether consciousness can occur when normal thoughts, memories, identity, and language are greatly reduced.

Sleep provides an interesting example for investigating this possibility. However, reports of experiencing darkness or a lack of thought after sleep cannot by themselves establish that consciousness continuously existed throughout the entire period.

Consequently, sleep is treated in this paper as a **question for investigation rather than proof**.

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4. The Brain–Consciousness Distinction

The central distinction proposed in this paper is:

> **The brain may generate and process information, while consciousness may be the system through which some of that information becomes subjectively experienced.**

The conventional scientific interpretation generally investigates consciousness as something arising from brain processes.

The alternative hypothesis proposed here asks whether the relationship could instead involve an additional direction of influence:

Brain → Conscious experience

versus

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Language and minimal mental content

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Claim: Conscious experience does not inherently require language.

Experiment G — Nonverbal perception task. Show consenting adult participants unfamiliar shapes, colours or sounds. Ask them to make nonverbal responses, such as matching, sorting or selecting a corresponding image, without naming the stimuli. Compare performance with a separate verbal-description condition. Successful perception and discrimination without verbal description supports the narrower claim that language is not required for these tasks. It does not show that language has no influence on thought.

Claim: Consciousness may occur when ordinary thoughts, identity, memory and language are reduced.

Experiment H — Carefully compare quiet wakeful states. Adult volunteers complete a brief eyes-open or eyes-closed quiet-rest session, followed by immediate reports of thoughts, imagery, sensations and awareness. Compare reports with non-invasive EEG if available through a qualified research setting. Do not use sleep deprivation, drugs or unsafe attempts to suppress thought. Reports of few thoughts cannot prove an absence of all mental content or establish content-free consciousness.

Claim: Sleep may help investigate minimal or absent mental content.

Experiment I — Use existing, ethically collected sleep-laboratory datasets or a qualified sleep laboratory. Compare EEG-defined sleep stages with immediate reports obtained after naturally occurring awakenings. Do not deliberately deprive participants of sleep. A report of no remembered experience is not evidence that no experience occurred; memory and reporting limits must be considered.

Ethics and safety: Human studies require informed consent, privacy protection, qualified adult supervision and review by an appropriate research ethics committee. These proposals are for adult volunteers in approved settings. Do not use sleep deprivation, drugs, anaesthesia, brain stimulation or invasive procedures as do-it-yourself experiments.

Brain ↔ Consciousness

The second model does not necessarily deny that brain activity is essential for consciousness. It proposes that the relationship may be reciprocal rather than entirely one-directional.

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5. The Primary Brain Hypothesis

If future experimental evidence demonstrated that consciousness can independently influence physical brain activity, then a new conceptual framework might be required.

This paper refers to that hypothetical system as the:

Primary Brain

The term does not refer to a literal second brain or an anatomical organ.

Instead, it describes a hypothetical **primary conscious system that could influence the biological brain**.

The proposed hierarchy would therefore be:

Consciousness / Primary System

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Biological Brain

↓

Neural processing

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Thoughts, memories, decisions, and actions

This terminology is proposed specifically for this hypothesis and should not be confused with established neuroscience terminology.

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6. The Energy and Activation Hypothesis

The strongest and most experimentally demanding part of the theory is the proposal that consciousness could somehow **activate, organize, or power brain activity**.

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Brain–consciousness distinction and reciprocal influence

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Claim: The brain may process information while consciousness is the system through which some information is experienced.

Experiment J — Awareness threshold study. Use a brief visual stimulus task with carefully controlled stimulus strength. Record stimulus parameters, eye movements, EEG and immediate awareness reports. Compare neural and behavioural patterns on reported-aware and reported-unaware trials. This can identify neural correlates of reported awareness, but cannot by itself establish whether the brain produces awareness or an additional system receives it.

Claim: The relationship might be reciprocal (brain ↔ consciousness), rather than one-directional.

Experiment K — Test temporal predictions, not just correlations. Preregister a model specifying which neural changes should precede awareness reports and which later neural or behavioural changes should follow them. Measure EEG, response timing and task performance in repeated trials. Compare the model with alternatives that explain both awareness and later activity through shared neural causes. Temporal order alone does not establish causal influence; a distinct, controlled causal prediction is required.

Ethics and safety: Human studies require informed consent, privacy protection, qualified adult supervision and review by an appropriate research ethics committee. These proposals are for adult volunteers in approved settings. Do not use sleep deprivation, drugs, anaesthesia, brain stimulation or invasive procedures as do-it-yourself experiments.

Experimental tests following original page 4

Primary Brain hypothesis and proposed hierarchy

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Claim: A hypothetical 'Primary Brain' or primary conscious system could influence the biological brain.

Experiment L — Competing-model test. Define 'Primary Brain' only as a proposed model, not an anatomical structure. Specify in advance one quantitative neural outcome that the extra-system model predicts and the conventional model does not. Collect non-invasive EEG and behavioural data in an ethically approved adult study; train models on one dataset and test them on an independent dataset. A replicated unexplained residual is a reason to improve measurement and compare alternative neural models, not evidence by itself for a Primary Brain. If no unique prediction can be specified, the proposal remains conceptual rather than experimentally falsifiable.

Claim: The proposed hierarchy is consciousness/primary system → biological brain → neural processing → thoughts, memories, decisions and actions.

Experiment M — Compare directional models. Use time-resolved recordings during a simple perception-and-response task. Compare preregistered models of information flow and their predictions on held-out trials. Existing neural signals can test information flow within measured brain networks. They cannot demonstrate a non-neural top-level system unless that system has an independently measurable variable and a unique prediction.

Ethics and safety: Human studies require informed consent, privacy protection, qualified adult supervision and review by an appropriate research ethics committee. These proposals are for adult volunteers in approved settings. Do not use sleep deprivation, drugs, anaesthesia, brain stimulation or invasive procedures as do-it-yourself experiments.

The phrase “power the brain” should not be interpreted to mean that consciousness has already been demonstrated to be a new form of energy.

The established biological brain requires energy supplied through metabolism. Neurons use electrochemical processes to communicate and maintain their activity.

The hypothesis proposed here is different:

> **If consciousness produces an independent and measurable physical influence on neural activity, then consciousness may represent an additional causal component of brain function.**

For this hypothesis to become scientifically meaningful, the influence would need to be measurable.

For example, researchers would need to determine whether conscious influence could produce a physical neural effect that cannot adequately be explained by known biological processes.

If no such effect exists, the “powering” component of the hypothesis would be weakened or rejected.

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7. Thoughts as Continuously Generated Information

Another component of the hypothesis concerns the nature of thoughts.

The brain can be conceptualized as a continuously changing information-processing system.

At any moment, numerous neural processes may be occurring simultaneously. Some information becomes relevant to conscious experience, while other information remains outside awareness.

A simplified conceptual model is:

Information generation

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Processing

↓

Selection

↓

Experimental tests following original page 5

Energy and activation hypothesis

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Claim: Consciousness may activate, organise or 'power' brain activity.

Experiment N — Require a physical mechanism and measurable quantity first. The paper correctly notes that the brain's established energy supply is metabolic and that consciousness has not been shown to be a new form of energy. Before testing, define exactly what 'activation' means (for example, a specified change in a neural signal) and how the proposed influence differs from ordinary neural, sensory, metabolic and attentional causes. Then compare preregistered models using non-invasive measurements. Do not describe an unexplained signal as new energy.

Claim: An independent conscious influence would produce a measurable neural effect not adequately explained by known mechanisms.

Experiment O — Out-of-sample prediction challenge. Build the strongest available conventional model using measured sensory input, task demands, attention indicators, eye movements and neural history. Specify the additional model's prediction before data collection, then test both on new participants or an independent dataset. Replication and successful exclusion of plausible confounds would strengthen the case for an unexplained causal factor. It still would not identify that factor as consciousness without an independent measurement and mechanism.

Falsification condition: If no independent effect is detected, the 'powering' component becomes less plausible.

Predefine a meaningful effect size and statistical sensitivity before the study. If the study is adequately powered, measurements are reliable, the predicted effect is absent and the confidence interval excludes the preregistered minimum effect, that specific version of the claim is weakened. A null result from an insensitive study is inconclusive.

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Experimental tests following original page 5

Thoughts as continuously generated information

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Claim: The brain can be modelled as a continuously changing information-processing system.

Experiment P — Continuous attention task. In a short computer task, present a changing sequence of neutral stimuli and ask participants to respond to occasional targets. Record reaction times, accuracy and, where available, EEG. Examine how processing changes over time and across task conditions. This tests dynamic information processing, not the full nature of thought.

Claim: Some information becomes relevant to awareness while other processing remains outside reported awareness.

Experiment Q — Masked versus clearly visible stimuli. Under an approved protocol, present brief visual targets at individually calibrated visibility levels, with appropriate masking and immediate awareness ratings. Compare objective discrimination, confidence and neural responses. Avoid claiming that a participant was definitely unconscious solely from a 'not seen' report; use converging measures and state the limits.

Claim: Thoughts may be temporary patterns rather than permanent objects.

Experiment R — Working-memory decay and updating. Ask adults to briefly remember neutral symbols, then introduce a distractor and test recall at different delays. Measure accuracy and interference. This can study the persistence and accessibility of information; it does not show that thoughts literally disappear or identify a consciousness-specific mechanism.

Ethics and safety: Human studies require informed consent, privacy protection, qualified adult supervision and review by an appropriate research ethics committee. These proposals are for adult volunteers in approved settings. Do not use sleep deprivation, drugs, anaesthesia, brain stimulation or invasive procedures as do-it-yourself experiments.

Experience / action / memory

↓

Storage or disappearance from accessible processing

In this model, thoughts are not necessarily permanent objects. They are temporary patterns of information processing.

This supports the idea that the conscious experience of a person may be a continuously changing stream rather than a sequence of permanently stored thoughts.

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8. Memory, Storage, and Deletion

The hypothesis proposes that information processed by the brain can have several possible outcomes.

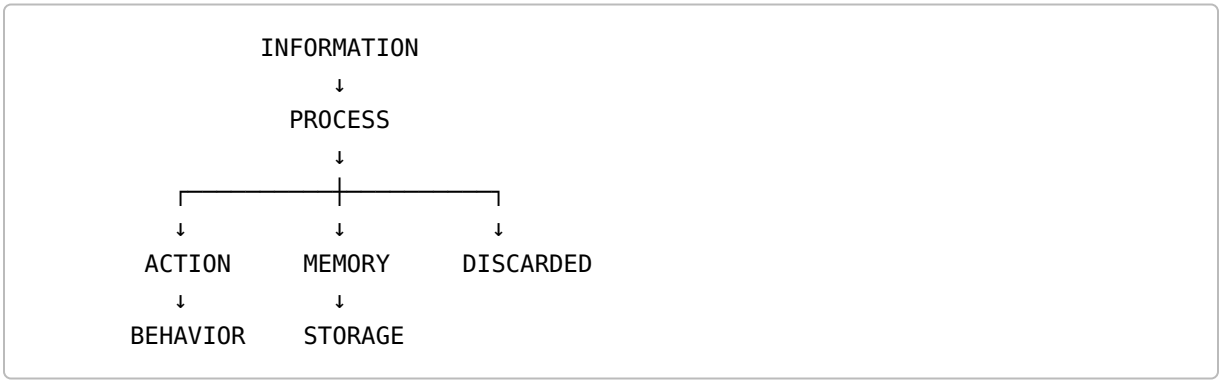
Information may be:

Executed
It influences an action or decision.

Stored
It becomes part of memory or another persistent neural representation.

Discarded
It ceases to remain accessible or important to subsequent processing.

This can be represented as:



This model does not claim that the brain literally labels each thought as “save” or “delete.” These terms are conceptual descriptions of different outcomes of information processing.

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Memory, storage, execution and discarding

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Claim: Processed information may guide action, be stored in memory, or become inaccessible.

Experiment S — Three-outcome memory task. Present neutral symbols linked to a simple response rule. Record which items guide an immediate action, which are recalled after a delay and which are not recalled. Include recognition tests to distinguish complete loss from failure of free recall. Analyse the outcomes separately. 'Discarded' should be treated as a convenient label for not being accessible in the chosen test, not proof that all traces were erased.

Claim: Information can influence behaviour without being consciously recalled.

Experiment T — Familiarity and memory test. Use a consented, low-risk computer task with neutral items, then compare free recall, recognition and performance on a later matching task. Any influence without free recall should be interpreted cautiously because implicit learning, familiarity and task strategy are alternative explanations.

Limitation: The brain does not literally label each thought 'save' or 'delete'.

Measurement note: Do not look for literal save/delete switches. Measure defined outcomes such as immediate response, delayed recall, recognition and later accessibility. State clearly which outcome each measure captures.

Ethics and safety: Human studies require informed consent, privacy protection, qualified adult supervision and review by an appropriate research ethics committee. These proposals are for adult volunteers in approved settings. Do not use sleep deprivation, drugs, anaesthesia, brain stimulation or invasive procedures as do-it-yourself experiments.

9. Consciousness and Sleep

Sleep creates an important problem for any theory of consciousness.

During sleep, the brain remains biologically active, but conscious experience can change dramatically depending on the sleep state.

A person may later report:

> “I experienced nothing.”

or

> “It felt like darkness.”

However, these reports must be interpreted carefully.

A memory of “nothing” does not necessarily demonstrate that a person continuously experienced nothing throughout sleep. The remembered representation may have been produced during waking or during a transition between sleep and wakefulness.

Therefore:

> **Sleep is an important thought experiment for the hypothesis, but it is not sufficient evidence that consciousness exists independently of the brain.**

This distinction prevents the hypothesis from relying on an unsupported conclusion.

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10. The Problem of Language

Language presents another useful distinction.

Human thought can involve language, but conscious experience does not appear to be limited to language.

A person can perceive an object before naming it.

A person can recognize a familiar face without verbally describing it.

A person can experience a sensation without possessing a word that completely describes it.

Therefore, this hypothesis treats language as a **representation system of the brain**, rather than as the substance of consciousness itself.

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Sleep and language claims

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Claim: Conscious experience changes across sleep states, and reports of 'nothing' do not establish continuous absence of experience.

Experiment U — Sleep-stage/report comparison. Use an existing sleep-lab dataset or work with a qualified sleep clinic and adult volunteers under ethics approval. Compare EEG-defined stages with immediate dream/experience reports after scheduled, gentle awakenings. Report the proportion of reports describing an experience, no remembered experience or uncertainty. The last category must not be treated as proof of unconsciousness.

Claim: A remembered representation of 'nothing' may arise at waking or during a transition.

Experiment V — Timing of reports. In an approved sleep study, compare reports collected immediately after awakening with reports collected after a short delay. Differences may indicate that recall and reporting timing affect descriptions. This cannot reconstruct an unremembered experience with certainty.

Claim: Language is a representation and communication tool, not necessarily the substance of consciousness.

Experiment W — Nonverbal recognition and naming. Present familiar objects or simple images and ask participants first to select a matching image, then separately to name or describe it. Compare accuracy and response time. Recognition without naming supports a distinction between some perceptual abilities and verbal labelling, not a complete theory of consciousness.

Ethics and safety: Human studies require informed consent, privacy protection, qualified adult supervision and review by an appropriate research ethics committee. These proposals are for adult volunteers in approved settings. Do not use sleep deprivation, drugs, anaesthesia, brain stimulation or invasive procedures as do-it-yourself experiments.

This supports the broader distinction between:

Consciousness

and

the information represented within consciousness.

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11. Predictions of the Hypothesis

A useful scientific hypothesis must make predictions.

If consciousness is an independent causal component of brain activity, then at least one measurable prediction should eventually distinguish this model from models in which brain activity completely explains consciousness.

Possible predictions include:

1. Consciousness should produce measurable physical effects on brain activity that cannot be explained by known neural mechanisms.

2. Changes in conscious state should produce specific causal changes in neural activity, rather than merely correlating with them.

3. If consciousness can influence the brain independently, controlled experiments should be able to detect this influence.

4. If no independent influence can ever be detected despite increasingly precise measurements, the “consciousness powers the brain” component of the hypothesis would become less plausible.

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12. Possible Experimental Approach

Testing the hypothesis would be extremely difficult because consciousness cannot currently be measured directly as an independent physical object.

However, research could investigate the relationship between:

subjective conscious state

and

measurable neural activity.

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Predictions and possible experimental approach

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Prediction: Consciousness should produce measurable physical effects not explained by known neural mechanisms.

Experiment X — Preregistered prediction test. The researchers must define the proposed independent variable, the neural outcome, the expected effect size and the conventional model before collecting data. Use non-invasive measures and an independent replication sample. If the hypothesis offers no measurable variable or unique prediction, this prediction cannot yet be tested.

Prediction: Changes in conscious state should cause specific neural changes, not merely correlate with them.

Experiment Y — Causal-inference study. Use only safe, approved manipulations of ordinary task conditions (such as stimulus visibility or attention instructions), while measuring neural activity and awareness reports. Compare causal models and control for sensory and attentional changes. Because the manipulation changes several brain processes, it cannot isolate consciousness as an independent cause by itself.

Prediction: Controlled experiments should detect an independent influence if one exists.

Experiment Z — Independent replication. Publish the protocol and analysis plan in advance, have a separate team repeat the same test, and report null as well as positive results. An effect that does not replicate should not be treated as established.

Prediction: Failure to detect an effect should weaken the claim.

Falsification plan: Define in advance what size of effect would matter, the required measurement quality and the number of observations. If well-powered replications consistently exclude that effect, reject or narrow that version of the hypothesis rather than changing the prediction after seeing the data.

Ethics and safety: Human studies require informed consent, privacy protection, qualified adult supervision and review by an appropriate research ethics committee. These proposals are for adult volunteers in approved settings. Do not use sleep deprivation, drugs, anaesthesia, brain stimulation or invasive procedures as do-it-yourself experiments.

Experiments could compare different states such as:

- normal wakefulness,
- different stages of sleep,
- anesthesia,
- dreaming,
- reduced sensory input,
- and other altered states of consciousness.

The important question would not simply be whether brain activity changes.

Instead, the question would be:

> **Does conscious experience have a measurable causal influence on brain activity beyond what existing physical models predict?**

This distinction between correlation and causation is essential.

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13. Contradictions and Alternative Explanations

The hypothesis has several major challenges.

13.1 Brain damage

Changes to the brain can alter memory, personality, perception, attention, and conscious experience.

This strongly demonstrates that consciousness is deeply connected to the physical brain.

However, it does not logically establish whether the brain is the complete source of consciousness or whether the brain is necessary for consciousness to function normally.

13.2 Sleep

The existence of unconscious or minimally conscious sleep states creates difficulty for the idea of continuous consciousness.

The hypothesis therefore cannot use subjective reports of sleep as definitive evidence.

13.3 Memory

If consciousness experiences something but the brain cannot encode the experience into memory, the person may later have no way to report it.

This creates an important limitation:

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Brain damage, sleep and memory: alternative explanations

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Claim: Brain damage changes memory, personality, perception, attention and conscious experience.

Evidence review rather than a new injury experiment: DEMOS should review published, ethically obtained clinical and neuroscience studies of naturally occurring brain lesions and neurological conditions. Compare which functions change with which brain networks are affected. Do not cause or seek brain injury. Such evidence demonstrates a strong dependence between brain function and conscious capacities, while the broader metaphysical question remains debated.

Claim: Sleep and memory make reports of experience difficult to interpret.

Experiment AA — Separate experience reports from memory performance. In safe waking tasks, compare immediate reports with delayed recall for the same neutral stimuli. This estimates how reporting delay affects memory. It cannot directly determine whether an unremembered experience occurred during sleep.

Ethics and safety: Human studies require informed consent, privacy protection, qualified adult supervision and review by an appropriate research ethics committee. These proposals are for adult volunteers in approved settings. Do not use sleep deprivation, drugs, anaesthesia, brain stimulation or invasive procedures as do-it-yourself experiments.

The absence of a memory cannot automatically prove the absence of consciousness.

At the same time, it also means that an unremembered conscious state is extremely difficult to investigate scientifically.

13.4 Existing neuroscience

Modern neuroscience provides extensive evidence connecting conscious states with brain activity.

Therefore, any theory proposing an independent consciousness must explain those findings rather than ignoring them.

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14. Limitations of the Hypothesis

This hypothesis currently has several limitations.

First, the concept of consciousness as a separate or primary system has not been experimentally established.

Second, the claim that consciousness “powers” the brain currently lacks a demonstrated physical mechanism.

Third, subjective experience is difficult to measure objectively.

Fourth, the term “Primary Brain” is a conceptual term introduced by this hypothesis rather than an established scientific term.

Finally, some observations used to motivate the hypothesis, particularly experiences associated with sleep, have alternative explanations.

These limitations do not automatically disprove the hypothesis. They identify the areas that require further investigation.

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15. Conclusion

This paper proposes that consciousness may be more than the collection of thoughts, memories, language, and sensory information produced by the brain.

The central hypothesis is that consciousness may function as an **observing and potentially activating system** associated with the biological brain.

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Existing neuroscience and limitations

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Claim: Existing neuroscience provides extensive evidence connecting conscious states with brain activity.

Evidence review: Summarise replicated findings from EEG, brain imaging, anaesthesia research and sleep research using peer-reviewed sources. For each finding, record what was measured, the sample, the result, limitations and whether it establishes correlation or causation. The current paper contains no bibliography, so references must be added before making specific literature claims.

Limitations: The separate-system concept, physical mechanism and objective measure are not yet established.

Research requirement: Define an independently measurable consciousness variable and a mechanism that produces a distinctive prediction. Until then, experiments can test awareness reports and neural correlates, but cannot directly confirm a separate Primary Brain.

Ethics and safety: Human studies require informed consent, privacy protection, qualified adult supervision and review by an appropriate research ethics committee. These proposals are for adult volunteers in approved settings. Do not use sleep deprivation, drugs, anaesthesia, brain stimulation or invasive procedures as do-it-yourself experiments.

The brain continuously processes information. Thoughts may be generated, selected, acted upon, stored, or discarded. Consciousness, according to the proposed model, is not identical to any single thought or memory but is associated with the experience of these processes.

The strongest part of the hypothesis is the proposal that consciousness may have a causal role in activating or organizing brain activity. If future experiments demonstrate an independent physical influence of consciousness on the brain, then consciousness could potentially be considered a primary component of brain function. This paper refers to that hypothetical concept as the **Primary Brain**.

However, this remains a hypothesis rather than an established scientific fact.

The central question therefore remains:

> **Is consciousness entirely produced by the physical brain, or does consciousness itself contribute causally to the activity of the brain?**

Answering that question would require evidence capable of distinguishing these possibilities experimentally.

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16. Future Research

Future investigation could focus on three major questions:

Question 1

Can consciousness exist without the normal contents of thought, language, memory, and sensory information?

Question 2

Can consciousness produce measurable physical effects on the brain?

Question 3

If such effects exist, can they be explained using existing neural and physical mechanisms?

If the answer to Question 2 were demonstrated experimentally and could not be explained by existing mechanisms, the proposed Primary Brain hypothesis would become substantially more scientifically significant.

Until then, the hypothesis should remain open to both confirmation and rejection.

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Conclusion and future research questions

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Claim: Consciousness may be more than the collection of thoughts, memories, language and sensory information.

Experiment AB — Factorial content-and-awareness study. Vary stimulus type and task demands while separately measuring awareness reports, content descriptions, memory and task performance. Test whether these measures are statistically separable. This can refine the conceptual distinction but cannot establish that consciousness exists independently of the brain.

Claim: Consciousness may have a causal role in activating or organising brain activity.

Experiment AC — Strongest available test. Define a mechanism and unique, preregistered prediction for the proposed influence; compare it with a comprehensive conventional neural model; measure relevant confounds; test on independent data; and require replication by another team. A result that merely correlates awareness reports with EEG is insufficient.

Future question: Can consciousness exist without normal thought, language, memory and sensory information?

Possible investigation: Study naturally occurring quiet-wakefulness reports and existing clinical/sleep datasets. Define each reduced-content condition separately. Do not infer content-free awareness from silence, stillness or absent recall. There is currently no accepted direct measurement that proves consciousness exists with absolutely no content.

Future question: Can consciousness produce measurable physical effects on the brain?

Possible investigation: Use Experiment N/O/AC above. This is the central unresolved test. The hypothesis must specify what effect would count, what conventional explanations must be excluded and what result would count against the claim.

Future question: If effects exist, can they be explained by existing neural and physical mechanisms?

Possible investigation: Compare the proposed model against the best available conventional explanations, including unmeasured neural activity, attention, sensory processing, expectation, memory, movement and measurement artefacts. Prefer the simpler model unless the additional model makes better replicated predictions.

Ethics and safety: Human studies require informed consent, privacy protection, qualified adult supervision and review by an appropriate research ethics committee. These proposals are for adult volunteers in approved settings. Do not use sleep deprivation, drugs, anaesthesia, brain stimulation or invasive procedures as do-it-yourself experiments.

Final Proposed Definition

Based on the ideas developed in this paper, the most scientifically careful definition would be:

> **Consciousness is hypothesized to be an observing capacity associated with subjective experience that may, in addition to receiving information from the biological brain, possess a causal influence on its physical activity. If such an independent causal influence were experimentally demonstrated, consciousness could be conceptualized as a primary system of brain function, referred to here as the "Primary Brain."**

Status of the proposal: Hypothesis / theoretical model, not an established scientific theory.

Experimental tests following original page 12

Final proposed definition and status

These are proposed ways to investigate the claims on the immediately preceding original page. They are not completed experiments or evidence. An unexplained result is not, by itself, proof of a separate consciousness system.

Claim: Consciousness may be an observing capacity associated with subjective experience and may possess a causal influence on physical brain activity.

Two-part test. Part 1: Investigate the observing-capacity definition with awareness reports, nonverbal discrimination and content measures (Experiments D–G and AB). Part 2: Investigate the proposed causal influence only after specifying a measurable mechanism and unique prediction (Experiments B, L, N/O and AC). Evidence for Part 1 does not automatically establish Part 2.

Claim: If an independent causal influence were demonstrated, consciousness could be conceptualised as a primary system of brain function called 'Primary Brain'.

Decision rule: Use the term 'Primary Brain' as a model label only if an independently measurable causal contribution is demonstrated, alternative explanations are rigorously tested, the prediction replicates independently and the mechanism is described. Until those conditions are met, retain it as a proposed concept, not an established brain system.

Status: Hypothesis / theoretical model, not an established scientific theory.

Scientific status note: This status should remain in the paper unless and until strong, independently replicated evidence justifies changing it. A proposed experiment, an unexplained result or a single positive study does not by itself establish the hypothesis.

Ethics and safety: Human studies require informed consent, privacy protection, qualified adult supervision and review by an appropriate research ethics committee. These proposals are for adult volunteers in approved settings. Do not use sleep deprivation, drugs, anaesthesia, brain stimulation or invasive procedures as do-it-yourself experiments.